

Effective and Enduring Place-Based Interventions Against Firearm Violence and Serious Crime Require Mechanism-Based Analytical Thinking

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Even in the face of recent decreases, community gun violence is a major public policy problem facing the United States. This problem occurs at the intersection of two enduring contexts: a national gun policy environment that offers little opportunity for affecting the purchasing and carrying of firearms—whether illegal or not—and an ongoing crisis of police legitimacy, which has dampened faith in enforcement-based strategies—and even increased purchasing and carrying of firearms.¹ As a consequence, interventions in the physical environment—which neither target guns nor involve the police—are attracting attention from researchers and policymakers.²

STREET CLEANING AND SERIOUS CRIME

In this issue of *AJPH*, MacDonald et al. (<https://bit.ly/3Qt4LFC>) continue their track record of rigorous field experiments on place-based interventions to reduce crime. This preregistered study

randomly assigned 300 street segments in Philadelphia, Pennsylvania, with the highest levels of gun violence to receive six months of cleanings bi-weekly (75) or monthly (75) or to act as untreated controls (150) with a random subset further being offered free trash and recycling bins. Treated segments received removal of sidewalk and street refuse, weeding, and planter maintenance, the efficacy of which was measured monthly by treatment-blinded observers using the Philadelphia Parks and Recreation department's own litter rating scale. This treatment mimics the real-world treatment of interest—and the measurement approach the city would likely use to assign and evaluate it—establishing the high external validity necessary to make the intervention policy relevant. The average treatment effect on police-reported gun crime was estimated using the canonical intent-to-treat difference-in-differences model. Across analyses, they accounted for spillover to nearby blocks, treatment heterogeneity, and treatment

compliance. This establishes the high internal validity necessary to be confident that observed impacts are attributable to the intervention.

The authors estimated a 16% to 17% reduction in monthly measured litter in both treatments. This reduction in observed litter, however, exerted no significant impact on gun violence in either the biweekly (incidence rate ratio [IRR] = 0.73; 95% confidence interval [CI] = 0.42, 1.28) or monthly treatments (IRR = 0.96; 95% CI = 0.45, 2.06). One limitation is that gun violence is thankfully uncommon and recently decreasing, resulting in wide CIs that do not rule out effects of cleaning on gun violence—but do suggest that if they exist, they are likely modest. Power is not easily purchased in ecologically valid field experiments; increasing the sample size would necessarily result in enrolling lower violence segments that are less relevant to the policy problem at hand. Despite this limitation, the implications are both clear and important: for policymakers, while simple cleaning programs are attractive to city governments because they are inexpensive and easy to implement—they use existing services and provide obvious direct benefits—they should not be expected to reduce gun violence. For researchers, it is further evidence against a “broken windows” direct effect of minor physical disorder on serious crime.³

ANALYTICAL THINKING FOR INTERVENTIONS AGAINST VIOLENCE

If cleaning programs offer limited promise, what other interventions in the physical environment of places may help prevent gun violence? Mechanism-based analytical theory and contextual

knowledge should be our guides. As MacDonald et al. write, “Public health and situational crime prevention approaches stress identifying the specific conditions driving violence concentration. These approaches focus on implementing tailored responses that address place characteristics that contribute to the dynamics of recurring violence” (p.xxx). Environmental interventions like the present study target community violence, which typically results from altercations associated with gang conflicts, illicit markets, and isolated interpersonal disputes where at least one party has a gun.

One way to intervene in community violence is to target gun availability (e.g., gun carrying) as it is a key proximal cause: without a gun, a shooting cannot occur. Most community violence is committed using handguns that are usually illegally owned or carried. Past work by MacDonald et al. have elsewhere demonstrated effectiveness of interventions that make lasting changes, such as turning vacant lots into parks and boarding up and beautifying abandoned buildings.² Their work suggests that reductions in gun crime occur in part because abandoned buildings and vacant lots in their setting are used to store firearms accessible during altercations but that would not be found if suspects were subject to a police frisk. Remediation programs thus align with the underlying problem: gun availability. However, remediation programs may not address shootings by lawful concealed carriers, from vehicles, or domestic violence—or if offenders are not concerned with being frisked by police.

Another way to intervene is to target altercations—for example, by increasing the number and willingness to intervene of bystanders who may

deter or de-escalate conflicts. Interventions into more transient conditions—such as trash removal—appear to impact crime mainly indirectly by increasing use of public spaces and interaction among residents, effects in common with more lasting blight remediation.⁴ Mechanistically, this may occur because interventions increase what criminologists call guardianship (“eyes on the street”) or collective efficacy (“informal social control”)³; they lead to more people watching and ready to deter or defuse altercations either directly or indirectly by calling the police. These effects may apply to a wider range of gun violence incidents (and crime more broadly) but are conditional on the presence of bystanders who believe that either direct intervention or calling the police will be effective in solving the problem and safe both for them—and even the subject(s) of the call. The conditions under which bystanders are willing to intervene and their interventions are effective are the subject of ongoing research.^{5,6}

DEVELOPING PLACE-BASED INTERVENTIONS THAT ENDURE

A final point is that both academics and policymakers often prefer interventions that have local and immediate effects because they are easily measured, are visible to the public, and can use existing city services. More invasive interventions into the built environment rarely have such isolated effects. For example, reducing automobility in favor of walkability, bicycling, and public transit has positive effects on crime including gun violence—but also injuries, physical and mental health, and the local economy.² Accurately measuring the full benefits from these interventions requires rigorous

evaluations with diverse (but preregistered) outcomes varying in follow-up that consider effects between and within individuals and at varying scales.

As I have argued elsewhere, lasting changes to the built environment—including deconcentration of public housing and changes to zoning⁷—provide long-term passive benefits.⁸ Rather than requiring upkeep—as in regular clean-up programs—they reduce the frequency of problems in the first place; a “good neighborhood” is one where many are willing to intervene but fortunately must do so only rarely. If built environment interventions increase perceived legitimacy and responsiveness of city government or increase monitoring of public spaces by filling them with people, they may have broad impacts on safety. These effects are diffuse and gradual but persistent, making neighborhoods and entire cities more stable, livable, and vibrant in the long term^{9,10}—but they are not amenable to what our disciplines recognize as the most convincing causal research designs.¹¹ While important for understanding how and when short-term interventions like street cleanings and blight remediation work, theory from the many disciplines that contribute to research and policy on community violence is even more important for designing policies with enduring benefits seen only long after grant funding windows or political terms. *AJPH*

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CONFLICTS OF INTEREST

The author declares no conflicts of interest.

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